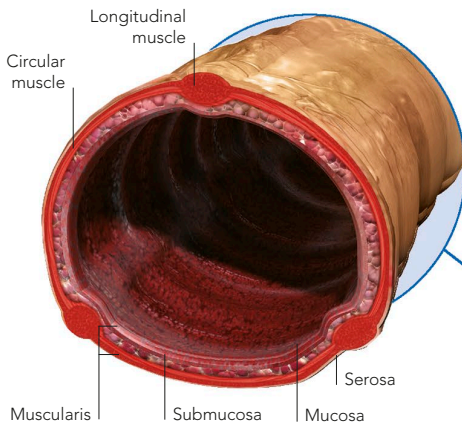


LARGE INTESTINE

THE LARGE INTESTINE HAS THREE MAIN REGIONS – THE CAECUM, COLON, AND RECTUM. THE 1.5M (5FT)-LONG COLON CHANGES LIQUID DIGESTIVE WASTE PRODUCTS INTO A MORE SOLID FORM THAT IS EXCRETED AS FAECES.

ROLE OF THE COLON

Once the chemical breakdown of food in the small intestine (see pp.218–19) is complete, almost all the nutrients vital for bodily functions will have been absorbed. The waste product from this process is partially digested, liquefied food (chyme). This passes from the small intestine, through the ileocaecal valve, into the caecum. From there, it reaches the first part of the colon, the ascending colon. The colon's main function is to convert the liquid chyme into semi-solid faeces for storage and disposal. Sodium, chloride, and water are absorbed through the lining of the colon into blood and lymph, and the faeces become less watery. The colon secretes bicarbonate and potassium in exchange for sodium and chloride. There are also billions of symbiotic or “friendly” microorganisms within the colon.



LAYERS OF THE COLON WALL

Inside the outer coating (serosa), smooth muscle fibres are responsible for colonic movements. The submucosa has many lymphoid nodules, and the mucosa produces lubricating mucus.

GUT FLORA

Trillions of microorganisms, mainly bacteria, live in the intestinal tract – chiefly in the large intestine. They are known as the gut flora (or gut microbiota) and have a vital role in human health and disease. They produce enzymes that break down certain food components, such as cellulose, which human enzymes cannot digest. The gut flora also produce vitamins K and B, and hydrogen, carbon dioxide, hydrogen sulphide, and methane. They help to control harmful microbes in the digestive system and promote the formation of antibodies and the activity of lymphoid tissue in the colonic lining. Overall, the gut flora and the body exist in a mutually beneficial partnership (symbiosis). At least one-third of the weight of excreted faeces is composed of these bacteria.

BACTERIA IN THE COLON

This electron microscope image (magnified over 2,000 times) shows clusters of rod-like bacteria (purple) on the lining of the colon.

